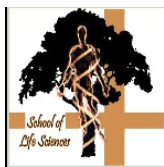


MANIPAL System Diagnostic Report



Molecular and Cellular Biology Unit

School of Life Sciences

Manipal University, Manipal - 576 104

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Email: dbt@manipal.edu

Report Date:

Friday, July 10, 2015 2:10:22 PM

Patient Name:

Specimen Type:

Gender:

Referring Physician:

Referring Department:

TEST CODE -

WIL006

CONDITION DIAGNOSED - Wilson's Disease

RESULT: Positive - A known variant causing the diagnosed condition was identified in the gene.

OBSERVED DNA VARIANTS:

1 Clinically significant mutation has been identified for the diagnosed condition.

Gene Name	Disease	Location of the Variant	Classification
ATP7B	Wilson's disease	chr13:52523808C>T c.2855G>A	Non-pathogenic

DETAILED VARIANT INFORMATION:

Gene Name	Variant Type	Variant Location	Amino Acid Change	dbSNP Identifier
ATP7B (RefSeq ID: NM_000053)	nonsynonymous SNV	exonic	p.R952K	rs732774

MINOR ALLELE FREQUENCY and PREDICTED IMPACT OF VARIANT:

MAF from 1000 Genomes Project	SIFT score and prediction	Polyphen score and prediction
0.53	0.99 and Tolerated	0.03 and Benign

ATP7B gene is a member of the P-type cation transport ATPase family and encodes a protein with several membrane-spanning domains, an ATPase consensus sequence, a hinge domain, a phosphorylation site, and at least 2 putative copper-binding sites. This protein functions as a monomer, exporting copper out of the cells, such as the efflux of hepatic copper into the bile. Alternate transcriptional splice variants, encoding different isoforms with distinct cellular localizations, have been characterized. Mutations in this gene have been associated with Wilson disease (WD). [provided by RefSeq, Jul 2008]

T C A T C A T G T C A C T T T G A C G T T G G T G G T A T G G A T T G T G A A T C G G T T T A T C
T C A T C A T G T C A C T T T G A C G T T G G T G G T A T G G A T T G T C A A T C G G T T T A T C

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